

Task 3: **Data Analysis Using Pandas**

Objective

Analyze a dataset, clean the data, and generate summary statistics and insights.

Requirements Completed

- ✓ Loaded CSV file using Pandas
- ✓ Performed data cleaning
- ✓ Applied filtering
- ✓ Performed grouping operations
- ✓ Generated summary statistics
- ✓ Produced meaningful insights

Code:

```
import pandas as pd

# Load CSV file
df = pd.read_csv("sales_data.csv")

print("Dataset Loaded Successfully\n")

# Show first rows
print(df.head())

# Check missing values
print("\nMissing Values:\n", df.isnull().sum())

# Remove duplicates
df.drop_duplicates(inplace=True)

# Convert date column
df["Date"] = pd.to_datetime(df["Date"])

# Create Total Sales column
df["Total_Sales"] = df["Quantity"] * df["Price"]

print("\nCleaned Data:\n", df)
```

```

# Summary statistics
print("\nSummary:\n", df.describe())

# Category wise sales
category_sales = df.groupby("Category")["Total_Sales"].sum()
print("\nCategory Sales:\n", category_sales)

# Best category
print("\nHighest Sales Category:", category_sales.idxmax())

# Total revenue
print("Total Revenue:", df["Total_Sales"].sum())

```

```

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# Category wise sales
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print("\nCategory Sales:\n", category_sales)

# Best category
print("\nHighest Sales Category:", category_sales.idxmax())

# Total revenue
print("Total Revenue:", df["Total_Sales"].sum())

```

```

25%    1002.750000    2.750000    2750.000000    2026-01-14 06:00:00    13500.000000
50%    1004.500000    3.500000    11000.000000    2026-01-19 00:00:00    37500.000000
75%    1006.250000    5.250000    26250.000000    2026-01-22 18:00:00    78750.000000
max     1008.000000    10.000000    55000.000000    2026-01-28 00:00:00    110000.000000
std       2.44949    2.66927    18761.796289          NaN    40746.603715

```

```

Category Sales:
Category
Clothing      20000
Electronics   335000
Furniture     29000
Name: Total_Sales, dtype: int64

```

```

Highest Sales Category: Electronics
Total Revenue: 384000

```

Dataset Loaded Successfully

	Order_ID	Customer	Category	Product	Quantity	Price	Date
0	1001	John	Electronics	Laptop	2	55000	2026-01-10
1	1002	Anu	Furniture	Chair	5	3000	2026-01-12
2	1003	Ravi	Electronics	Phone	3	25000	2026-01-15
3	1004	Meena	Clothing	Shirt	10	800	2026-01-18
4	1005	Arun	Furniture	Table	2	7000	2026-01-20

```

Missing Values:
Order_ID      0
Customer      0
Category      0
Product       0
Quantity      0
Price         0
Date          0
dtype: int64

```

Cleaned Data:

	Order_ID	Customer	Category	...	Price	Date	Total_Sales
0	1001	John	Electronics	...	55000	2026-01-10	110000
1	1002	Anu	Furniture	...	3000	2026-01-12	15000
2	1003	Ravi	Electronics	...	25000	2026-01-15	75000
3	1004	Meena	Clothing	...	800	2026-01-18	8000
4	1005	Arun	Furniture	...	7000	2026-01-20	14000
5	1006	Kavi	Electronics	...	15000	2026-01-22	60000
6	1007	Latha	Clothing	...	2000	2026-01-25	12000
7	1008	Rahul	Electronics	...	30000	2026-01-28	90000

[8 rows x 8 columns]

Summary:

	Order_ID	Quantity	Price	Date	Total_Sales
count	8.000000	8.000000	8.000000	8	8.000000
mean	1004.500000	4.375000	17225.000000	2026-01-18 18:00:00	48000.000000
min	1001.000000	2.000000	800.000000	2026-01-10 00:00:00	8000.000000

Conclusion:

The dataset was successfully cleaned and analyzed using Pandas, and meaningful insights were generated from the data.